

Lecture 9

Particle Dynamics

Art and Lenny expected some action at Hilbert's Place. But all the state-vectors were absolutely still—frozen, you might say.

Lenny: *This is boring, Art. Doesn't anything ever happen around here? Hey Hilbert, why is this joint so still?*

Hilbert: *Oh, don't worry. Things will pick up as soon as the Hamiltonian gets here.*

Art: *The Hamiltonian? He sounds like a real operator.*

9.1 A SIMPLE EXAMPLE

The first two volumes of the Theoretical Minimum series have largely focused on two questions. The first is: What do we mean by a system and how do we describe the momentary states of a system? As we've seen, the classical and quantum answers to this question are very different. Classical phase space—the space of coordinates and momenta—is replaced in quantum theory by the linear vector space of states.

The second big question is: How do states change with time? In both classical mechanics and quantum mechanics, the answer is *according to the*